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SUPERSEDING
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FEDERAL SPECIFICATION

STEEL, ALLOY, SHEET AND STRIP, HOT-ROLLED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers hot-rolled alloy-steel sheet and strip used for general fabrication purposes.

1.2 Classification. — Steel sheet and strip shall be classified in the following manner (see 6.1.2).

1.2.1 *Steel-grade numbers.* — Steel sheet and strip is furnished in the following standard alloy-steel numbers, as specified (see 6.1):

4130	5140	8620
4135	5150	8630
4140	5160	8640
4142	6150	8645
4145	8615	8735
4150	8617	9262
4340		

1.2.2 *Conditions.* — Steel sheet and strip shall be furnished in the following conditions, as specified (see 6.1):

HR—As hot-rolled.

HRA—Hot-rolled and annealed.

HRSA — Hot-rolled and spheroidize annealed.

HRN—Hot-rolled and normalized.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 48—Tolerances for Steel and Iron Wrought Products.

Fed. Std. No. 66—Steel: Chemical Composition and Hardenability.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the

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Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-163 — Steel Mill Products, Preparation for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition.—Chemical composition of steel sheet and strip shall be ordered to the steel designation numbers in accordance with 1.2.1. The composition of the grade designation numbers shall be in accordance with Federal Standard No. 66 (see 6.1). A ladle analysis of each heat of steel shall be furnished by the supplier.

3.1.1 *Check analysis.*—The chemical composition, as determined by check analysis, shall meet the requirements for the specified grade designation provided that in a heat of steel the individual tolerances do not vary from above and below the ranges shown for check analysis in Federal Standard No. 66.

3.2 Mechanical properties.

3.2.1 *Tensile requirements.*—The specimens representing the steel sheet or strip shall meet the tensile requirements when these requirements are specified in the invitation for bids, contract, or order (see 6.1).

3.2.2 *Cold-bending requirements.*

3.2.2.1 *Spheroidize-annealed condition.*—Unless otherwise specified in the invitation for bids, contract or order, the specimens representing steel sheet and strip specified in the spheroidize-annealed condition shall meet without cracking the bend requirements of table I.

3.2.2.2 *Annealed and normalized condition.*—If cold-bending requirements are specified in the invitation for bids, contract, or order for steel sheet or strip specified in the hot-rolled annealed or hot-rolled normalized condition, the test requirement shall be negotiated between the supplier and procuring agency.

3.2.2.3 *As hot-rolled condition.*—Bend-test requirements shall not apply to steel sheet and strip in the as-hot-rolled condition.

3.3 Spheroidization. — Unless otherwise specified in the invitation for bids, contract, or order, steel sheet and strip specified in the spheroidize-annealed condition shall be as least 75 percent spherodized. When other than the above percentage of spheroidization

TABLE I.—*Bend requirements for sheet and strip (spheroidize-annealed condition)*

Thickness	Carbon content, percent	Degree of bend	Ratio of bend radius to thickness of specimen	Relation of bend to rolling direction
All	Up to 0.30, incl.	180	½t	Perpendicular
⅛ inch and under	Over 0.30	180	½t	Perpendicular
Over ⅛ inch	Over 0.30	180	t	Perpendicular

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is required, the standards shall be mutually agreed upon between the supplier and the procuring agency (see 6.1).

3.4 Finish.—Unless otherwise specified in the invitation for bids, contract, or order, steel sheet and strip shall be furnished with a pickled finish. The surface of the sheet or strip shall be free of scale. Unless otherwise specified, all pickled sheet shall be coated with a preservative oil. Sheet and strip which are ordered with an as-hot-rolled or as-heat-treated finish shall be furnished dry.

3.5 Edges.—Unless otherwise specified in the invitation for bids, contract, or order, steel sheet shall be furnished with sheared edges, and steel strip shall be furnished with either mill edges or slit edges, subject to agreement between the supplier and the procuring agency.

3.6 Dimensions. — Hot-rolled alloy-steel sheet or strip shall be furnished in the sizes (see 6.1) as specified in the invitation for bids, contract, or order. Tolerances from specified dimensions shall conform to Federal Standard No. 48. The applicable references to this standard are listed in table II.

TABLE II.—*Dimensional tolerance reference to Fed. Std. No. 48*

Dimensional tolerance	Fed. Std. No. 48 paragraph reference	
	Sheet	Strip
Thickness	11b1	14b1
Crown	----	14b2
Width	11b2	14b3
Length	11b3	14b4
Camber	11b7	11b5
Flatness	11b8	14b6
Out-of-square	11a9	11a9
Resquared sheet	11a10	----

3.7 Identification marking. — Identification marking shall be as specified in the invitation for bids, contract or order.

3.8 Workmanship.—The hot-rolled alloy-steel sheet and strip shall be clean and free of imperfections, such as laminations, segregation, and surface defects, as is consistent with good commercial practice.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements prior to the submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be complete and available to the Government as specified in the contract or order.

4.2 Lot.—Unless otherwise specified, a lot shall consist of all sheet or strip submitted for inspection at one time, of the same heat, the same condition and finish, the same thickness and, when heat treated, subjected to the same heat treatment.

4.3 Sampling.

4.3.1 *For chemical composition.*—At least one sample shall be taken from each lot for check analysis in accordance with methods 111 or 112 of Federal Test Method Standard No. 151.

4.3.2 *Mechanical properties.*

4.3.2.1 *Tensile.* — When tensile requirements are specified, at least one test shall be made for each 10,000 pounds of material or fraction thereof in each lot.

4.3.2.2 *Cold bending.* — One sample for bend-test requirements for sheet or strip ordered in the spheroidize-annealed condition shall be taken from each lot. When the supplier and the procuring agency agree upon bend-test requirements for hot-rolled annealed or hot-rolled normalized material and the requirements are specified in the contract or order, one sample shall be taken from each lot.

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4.3.3 *Spheroidization.*—At least one sample shall be taken from each lot ordered in the spheroidize-annealed condition.

4.4 Examination.

4.4.1 *Visual.*—A representative sampling of the sheet and strip in each lot shall be examined for compliance with the requirements for finish (see 3.4), edge (see 3.5) and workmanship (see 3.7).

4.4.2 *Dimensional.*—A representative number of measurements shall be taken in each lot to determine compliance with size and dimensional tolerances (see 3.6). Thickness measurements shall be made with a micrometer.

4.4.3 *Preparation for delivery.*—Prior to shipment, examination shall be made to determine compliance with the requirements of section 5.

4.5 Tests.

4.5.1 *Test specimens.*

4.5.1.1 *Chemical composition.*—Specimens for chemical analysis shall be prepared in accordance with methods 111 or 112 of Federal Test Method Standard No. 151 for the applicable procedure.

4.5.1.2 *Mechanical properties.*

4.5.1.2.1 *Tensile.* — Specimens for tensile tests shall be prepared in accordance with method 211 of Federal Test Method Standard No. 151 and shall conform to the shapes and dimensions of type F2 or F3 test specimens.

4.5.1.2.2 *Cold bending.* — Specimens for cold bending shall be prepared in accordance with method 231 of Federal Test Method Standard No. 151.

4.5.1.3 *Spheroidization.* — Specimens for microexamination for spheroidization shall have a prepared surface at least one inch long and shall represent the full thickness of the material.

4.5.2 *Test Methods.*

4.5.2.1 *Chemical composition.* — Chemical analysis shall be conducted in accordance with methods 111 or 112 of Federal Test Method Standard No. 151.

4.5.2.2 *Mechanical properties.*

4.5.2.2.1 *Tensile.* — Tensile tests shall be conducted in accordance with method 211 of Federal Test Method Standard No. 151.

4.5.2.2.2 *Cold bending.*—Cold-bending test shall be conducted in accordance with method 231 of Federal Test Method Standard No. 151.

4.5.2.3 *Spheroidization.* — Spheroidization tests shall be conducted at sufficient magnification to give reliable results. When standards are agreed upon between the supplier and procuring agency the results of the test shall be compared against these standards.

4.6 Rejection.

4.6.1 *Examination.*—If the representative sampling for visual, dimensional, or preparation for delivery fails to meet the requirements of the specification when examined in accordance with 4.4, the lot shall be rejected.

4.6.2 *Tests.*—A lot shall be rejected for failure to comply with any of the specified test requirements when tested in accordance with 4.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing (Civil agencies). — Definitions and applications of the levels of preservation, packaging, and packing shall be in accordance with Federal Standard No. 102.

5.2 Preservation.—Sheet and strip shall be prepared for shipment in accordance with levels A, B, or C as specified (see 6.1).

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5.2.1 *Levels A and B.*—Preservation for shipment shall be in accordance with Military Standard MIL–STD–163.

5.2.2 *Level C.*—Preservation for shipment shall be in accordance with commercial practice.

5.3 Packing. — Sheet and strip shall be packed for shipment in accordance with levels A, B, or C, as specified (see 6.1).

5.3.1 *Levels A and B.*—Packing for shipment shall be in accordance with Military Standard MIL–STD–163.

5.3.2 *Level C.*—Packing for shipment shall be in accordance with commercial practice adequate to ensure carrier acceptance and safe delivery at the lowest rates.

5.4 Marking for shipment.

5.4.1 *Civil agencies.* — Marking for shipment shall be in accordance with Federal Standard No. 123.

5.4.2 *Military agencies.*—Marking for shipment shall be in accordance with Military Standard MIL–STD–163.

6. NOTES

6.1 Ordering data. — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Steel grade designations (see 1.2.1).
(*c*) Condition (see 1.2.2).
(*d*) Mechanical properties (see 3.2).
(*e*) If cold bending is required for annealed or normalized condition (see 3.2.2.2).
(*f*) If special spheroidization requirements are required (see 3.3).
(*g*) If other than pickled finish is required (see 3.4).
(*h*) If special edges are required (see 3.5).
(*i*) Size and dimensions (see 3.6).
(*j*) Identification marking required (see 3.7).
(*k*) If special lot size is required (see 4.2).
(*l*) Level of preservation and packing required (see 5.2 and 5.3).

6.1.1 *Method of specifying thickness.*—The thickness of sheet and strip should be specified in decimals of an inch rather than in terms of gage numbers.

6.1.2 *Classification of sheet and strip.*—Sheet and strip is classified as shown in table III.

TABLE III.—*Hot-rolled, flat-rolled alloy steel classified as sheet or strip according to size*

Width, inch	Thickness, inch					
	0.2299 to 0.2031	0.2030 to 0.1800	0.1799 to 0.0568	0.0567 to 0.0344	0.0343 to 0.0255	under 0.0255
To 3.5, incl.	[1]	Strip	Strip	Strip	Strip	Sheet
Over 3.5 to 6.0 incl.	[1]	Strip	Strip	Strip	Sheet	Sheet
Over 6.0 to 24.0 excl.	Strip	Strip	Strip	Sheet	Sheet	Sheet
24.0 to 48.0, incl.	Sheet	Sheet	Sheet	Sheet	Sheet	Sheet
Over 48.0	[1]	[1]	Sheet	Sheet	Sheet	Sheet

[1] Not applicable to sheet or strip.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEM PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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